

CYBERSECURITY - Week 1

Threat and Vulnerability Identification and Basic Security Assessment

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Basic Security Assessment

Whois

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What is a Threat ?

A threat is anything that can cause damage to a system.

Examples:

- Hacker
- Malware
- Virus
- Phishing email
- Unauthorized access

TYPES OF THREAD MALWARE

❑ Malware (Malicious Software)

- Malware refers to malicious software designed to harm or exploit devices, networks, or services.

Examples:

- **Viruses:** Infect files or systems and spread across devices.
- **Worms:** Self-replicating programs that spread through networks.
- **Trojan Horses:** Malicious programs disguised as legitimate software.
- **Ransomware:** Encrypts data and demands payment for its release.
- **Spyware:** Collects sensitive information without user consent.
- **Adware:** Displays unwanted advertisements and may install malicious software. It can appear as banner ads, pop-up ads in your computer or mobile device.

TYPES OF THREAD

- ❑ **Phishing:** Deceptive emails or messages designed to steal credentials or personal data.
- **Vishing** (voice phishing): It is a type of phishing attack in which fraudulent phone calls or voice messages designed to trick victims into providing sensitive information, like login credentials, credit card numbers, or bank details.
- **Smishing:** It is a term used to describe phishing attacks that occur via **SMS (Short Message Service)**, also known as **SMS phishing**. It involves attackers sending fraudulent text messages to trick individuals into revealing sensitive information, such as login credentials, personal identification numbers (PINs), or financial data.
- ❑ **DoS:** denial-of-service (DoS) attack in which a single computer or device sends a large number of requests to a target, making it unavailable to users.
- ❑ **DDoS:** A distributed denial-of-service (DDoS) attack in which Multiple compromised systems send a large number of requests to a target, making it unavailable to users. .

What is a Vulnerability ?

A vulnerability is a weakness that can be exploited by a threat.

These weaknesses can be exploited by attackers to gain unauthorized access, steal data, or use the system for malicious purposes.

Examples:

- Open ports
- Weak passwords
- Outdated software
- Unencrypted communication

Types of Vulnerabilities

- Configuration Issues – Weak passwords, misconfigured firewalls.
- Software Bugs – Unpatched applications, outdated OS.
- Network Weaknesses – Open ports, insecure protocols.
- Human Errors

Security vulnerabilities and their impact

❑ Code Vulnerabilities:

- **Buffer Overflow:** Occurs when a program writes more data to a buffer than it can hold, potentially allowing attackers to execute arbitrary code.
- **SQL Injection:** Malicious SQL statements are inserted into a query to manipulate databases.
- **Cross-Site Scripting (XSS):** Attackers inject malicious scripts into web pages viewed by others.

❑ Configuration Issues:

- **Default Credentials:** Using default usernames and passwords, which attackers can easily guess.
- **Misconfigured Access Controls:** Allowing unauthorized users to access sensitive data or functionalities.
- **Exposed APIs:** APIs without proper authentication or authorization controls.
- **Outdated Systems:** Using unpatched or outdated software that contains known vulnerabilities.

❑ Human Factors:

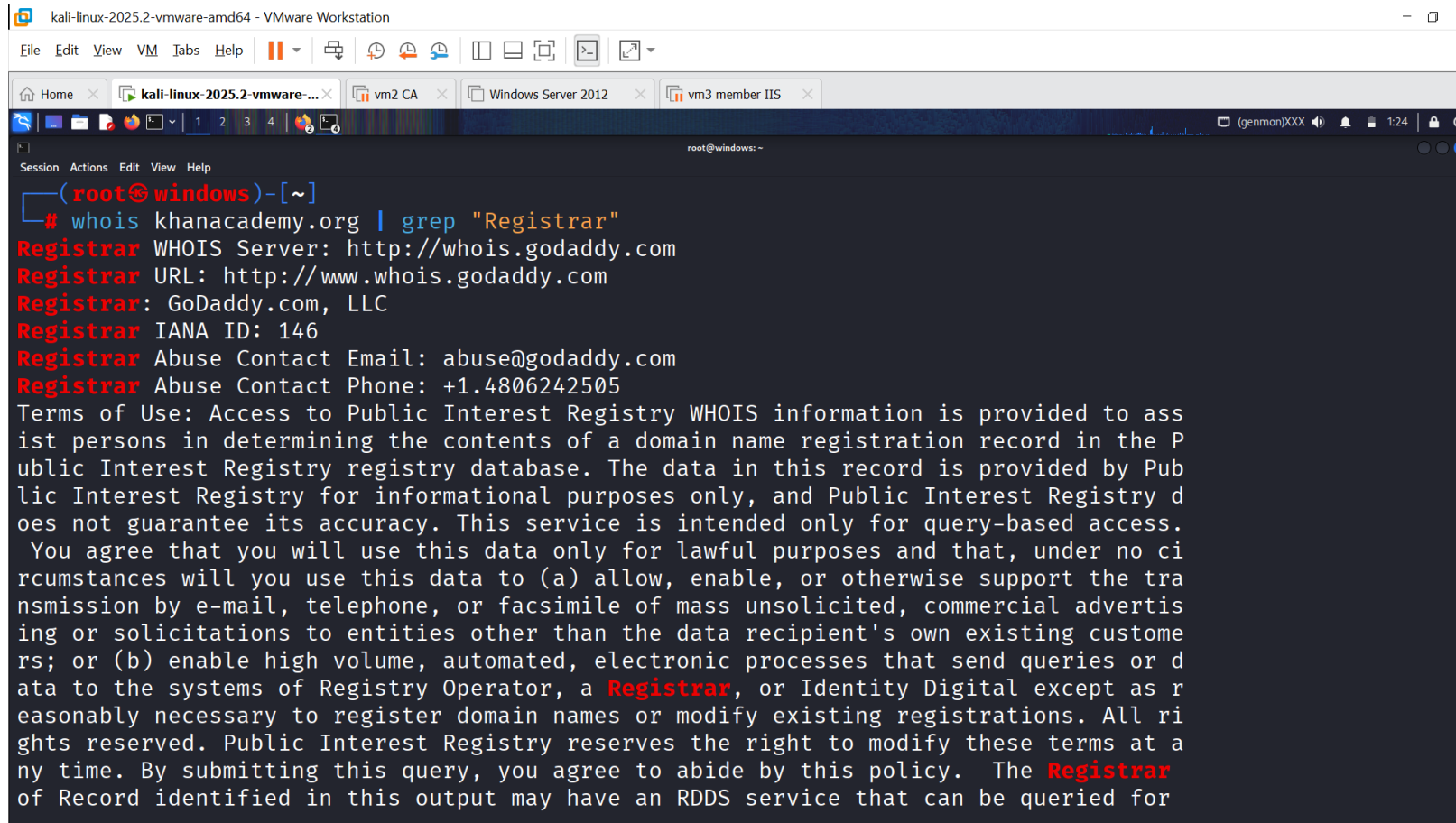
- **Phishing Attacks:** Tricking individuals into revealing sensitive information.
- **Social Engineering:** Manipulating people to gain access to systems or data.

Steps in Vulnerability Testing :

Planning – Define the scope (systems, apps, networks).

- **Scanning** – Use tools (like Nessus, OpenVAS, Nikto) to find vulnerabilities.
- **Analysis** – Evaluate the risk level (Low, Medium, High, Critical).
- **Reporting** – Create a detailed report for IT/security teams.
- **Remediation** – Apply patches, fix misconfigurations, update systems.
- **Verification** – Re-scan to ensure vulnerabilities are resolved.

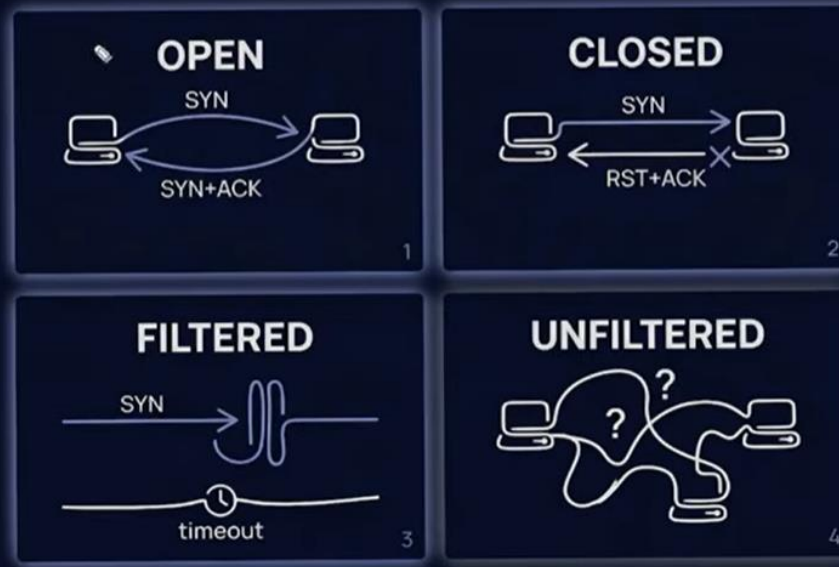
Whois cmd – is use to check the domain owner, registration/expiry dates, registrar, nameservers, and domain status to know who owns it and how the domain is managed.



```
kali-linux-2025.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.2-vmware-... vm2 CA Windows Server 2012 vm3 member IIS
root@windows: ~
Session Actions Edit View Help
(root@windows)-[~]
# whois khanacademy.org | grep "Registrar"
Registrar WHOIS Server: http://whois.godaddy.com
Registrar URL: http://www.whois.godaddy.com
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of a domain name registration record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only, and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, telephone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b) enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to modify these terms at any time. By submitting this query, you agree to abide by this policy. The Registrar of Record identified in this output may have an RDDS service that can be queried for
```

Port states

Port States — What NMAP Sees



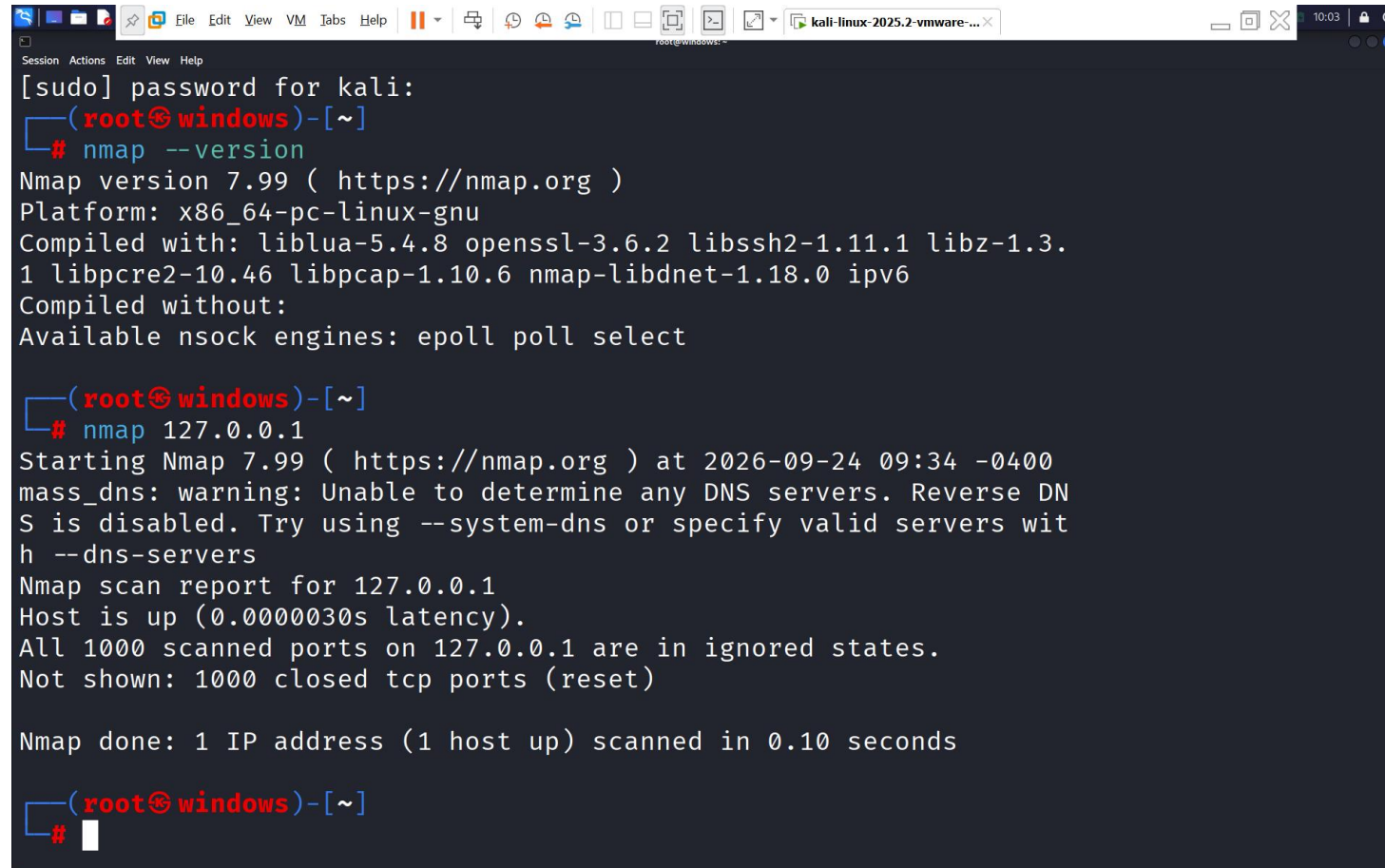
How NMAP Interprets Responses

- **Open:** SYN → SYN+ACK returned. App is listening.
- **Closed:** SYN → RST+ACK returned. Host up, no app.
- **Filtered:** SYN → no response (dropped). Firewall active.
- **Unfiltered:** Reachable, but state indeterminate.
- **open|filtered / closed|filtered:** Ambiguous — no conclusive response received.

nmap

- **Nmap (Network Mapper)** is a **network scanning tool used to discover devices, open ports, potential vulnerabilities and services on a network.**
- By default scans 1,000 common TCP ports.
- Host , port scan
- Tcp ,udp connect scan

Nmap practical



The screenshot shows a terminal window with a dark background. The window title bar includes icons for file operations and a tab labeled 'kali-linux-2025.2-vmware-...'. The terminal content shows a user at the root of a 'windows' machine. They first run 'sudo' and enter the password for 'kali'. Then they run 'nmap --version', which displays the Nmap version (7.99), platform (x86_64-pc-linux-gnu), and compilation details. Next, they run 'nmap 127.0.0.1', which shows a scan report for the local host, indicating it is up and that all 1000 scanned ports are in ignored states (closed TCP ports).

```
[sudo] password for kali:
(root@windows)-[~]
# nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.
1 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(root@windows)-[~]
# nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 09:34 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DN
S is disabled. Try using --system-dns or specify valid servers wit
h --dns-servers
Nmap scan report for 127.0.0.1
Host is up (0.0000030s latency).
All 1000 scanned ports on 127.0.0.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

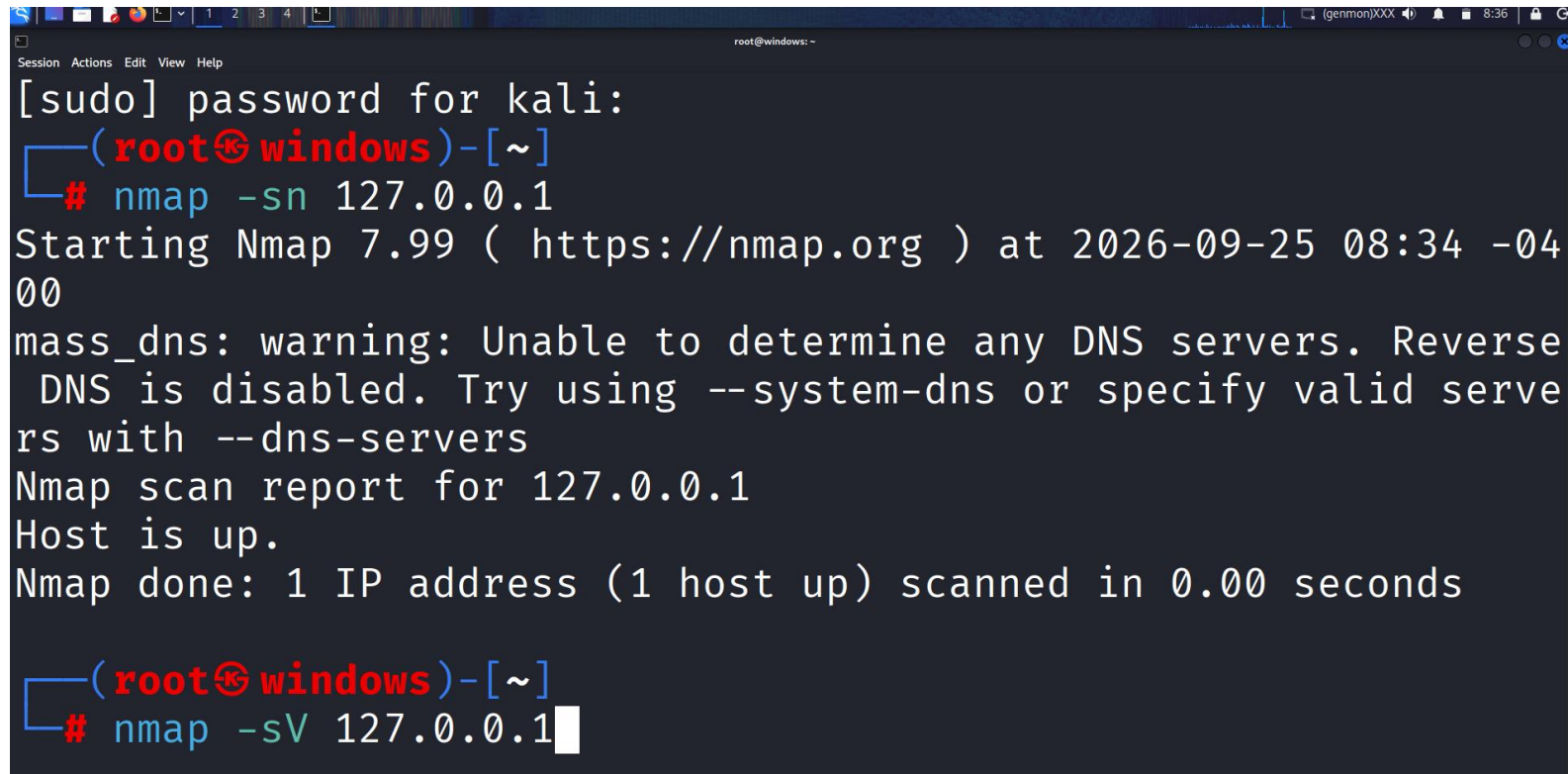
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

(root@windows)-[~]
#
```

Host Discovery

–sn : The -sn is used to detect live hosts without scanning for open ports.

- **Syntax:** nmap –sn <ip address>

A terminal window titled 'root@windows: ~' showing the execution of the nmap -sn command. The prompt is '(root@windows)-[~]'. The user enters '# nmap -sn 127.0.0.1'. The terminal output shows 'Starting Nmap 7.99 (https://nmap.org) at 2026-09-25 08:34 -0400', a warning about DNS, the scan report for 127.0.0.1 stating 'Host is up.', and 'Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds'. The prompt returns to '(root@windows)-[~]'. The user then enters '# nmap -sV 127.0.0.1' and the cursor is at the end of the command.

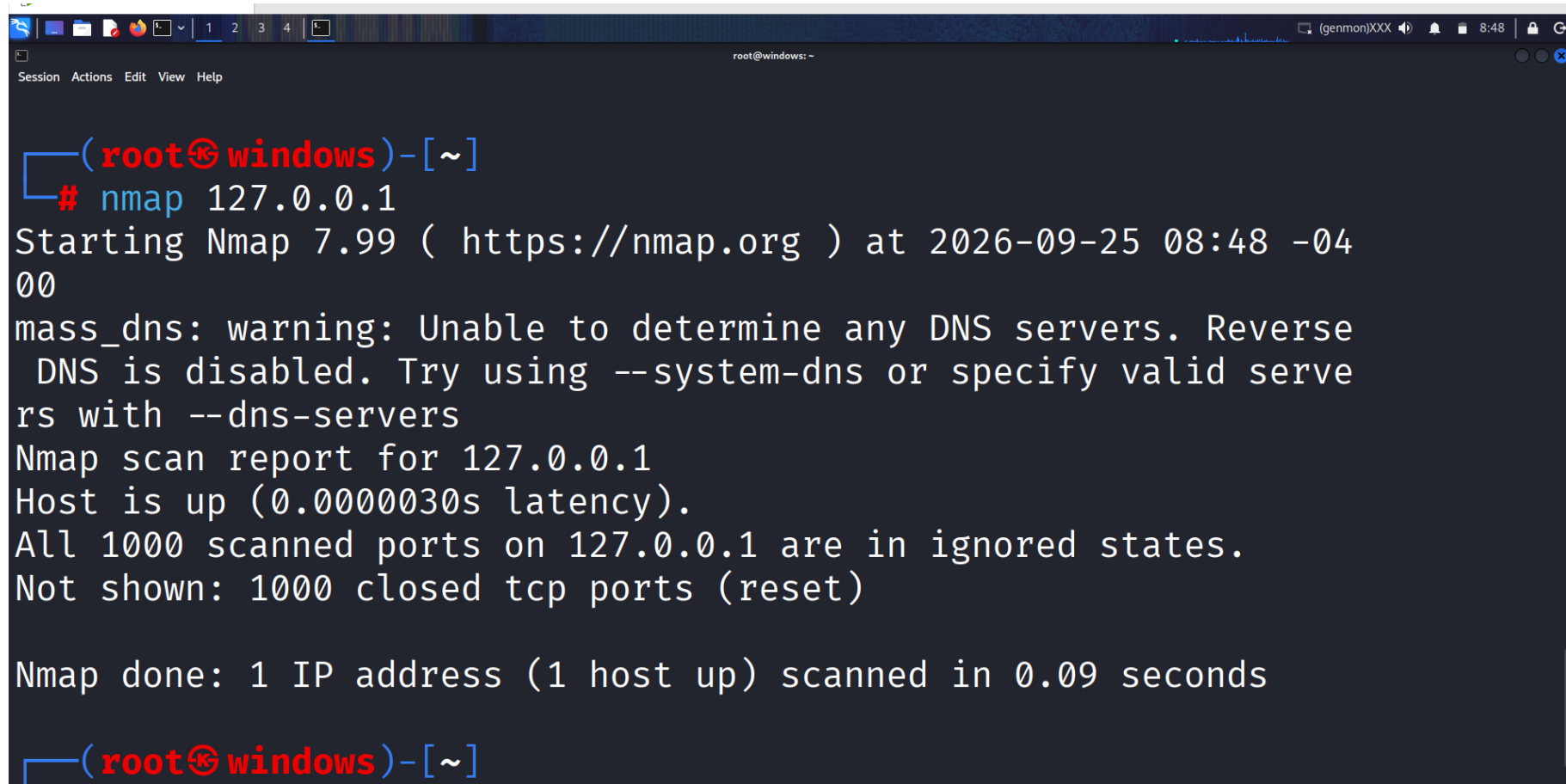
```
Session Actions Edit View Help
root@windows: ~
[sudo] password for kali:
(root@windows)-[~]
# nmap -sn 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 08:34 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse
DNS is disabled. Try using --system-dns or specify valid serve
rs with --dns-servers
Nmap scan report for 127.0.0.1
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds

(root@windows)-[~]
# nmap -sV 127.0.0.1
```

Port scanning

To identify open ports on the target system.

During security assessments to find exposed services.

A screenshot of a terminal window titled 'root@windows: ~'. The terminal shows a command prompt where the user has entered 'nmap 127.0.0.1'. The output of the command is displayed, including the Nmap version (7.99), the scan date and time (2026-09-25 08:48 -0400), a warning about DNS, the scan report for 127.0.0.1, and the final result indicating that all 1000 scanned ports are in ignored states (reset). The terminal window has a dark background and a menu bar at the top with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The system tray at the top right shows the user 'genmon)XXX', a volume icon, a notification icon, and the time '8:48'.

```
(root@windows)-[~]
# nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 08:48 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse
DNS is disabled. Try using --system-dns or specify valid serve
rs with --dns-servers
Nmap scan report for 127.0.0.1
Host is up (0.0000030s latency).
All 1000 scanned ports on 127.0.0.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

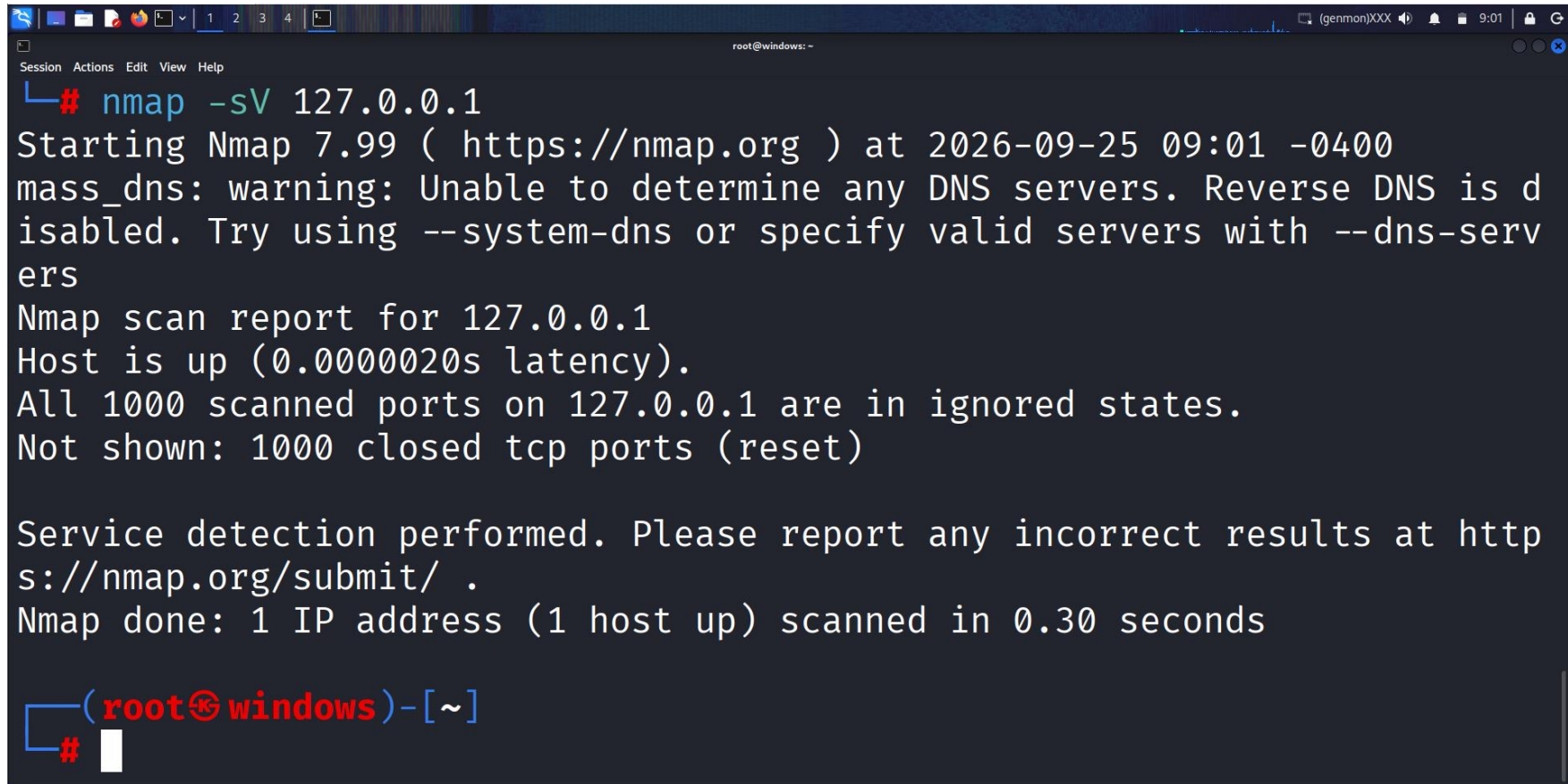
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds

(root@windows)-[~]
```

Service Version Detection

-sV: It helps identify **which services** are running on open ports and **what versions** they are using .

Syntax: `nmap -sV <ip address>`

A terminal window titled 'root@windows: ~' showing the execution of the nmap -sV 127.0.0.1 command. The output includes the Nmap version (7.99), a warning about DNS, the scan report for 127.0.0.1, and the final status indicating 1 IP address scanned in 0.30 seconds. The prompt at the bottom is '(root@windows)-[~]' with a cursor.

```
root@windows: ~  
# nmap -sV 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 09:01 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 127.0.0.1  
Host is up (0.0000020s latency).  
All 1000 scanned ports on 127.0.0.1 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds  
  
(root@windows)-[~]  
#
```

OS Detection

To identify the operating system of the target host.



```
(root@windows)-[~]
# nmap -O 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 09:03 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 127.0.0.1
Host is up (0.000081s latency).
All 1000 scanned ports on 127.0.0.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds

(root@windows)-[~]
#
```